

Offshore Wind Farm Profitability Analysis

Data-Driven CAPEX Estimation & Decision Support

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Motivation

- Offshore wind is a cornerstone of the energy transition – expected to deliver 30%+ of European electricity by 2040.
- Projects require multi-billion EUR investments; accurate CAPEX estimation is critical for feasibility decisions.
- No single open-source tool integrates historical project data, environmental parameters, and predictive modelling.

Goals

- 1 Predict unit CAPEX (€/MW) for new offshore wind projects.
- 2 Identify the most important features driving capital costs.
- 3 Deliver an interactive decision-support tool (web UI).

Input Raw Data

10 original columns:

- wind_farm_name
- country
- commissioning_year
- installed_capacity_MW
- turbine_model
- foundation_type
- water_depth_m
- distance_from_shore_km
- total_project_budget
- project_lifetime_years

Problems

- Only **182 records** – very small dataset for machine learning.
- **Lots of missing data** in critical columns (total_project_budget, installed_capacity_MW).
- Budgets encoded as free text in mixed currencies: *“GBP 4.2 billion”*, *“USD 3.64 billion”*.
- No geographic coordinates (LAT, LON) or farm area.

Manual Data Filling

Several critical columns were **not present in the raw dataset** and had to be filled through manual web research:

Columns added manually

- LAT, LON – geographic coordinates
- area_sqkm – farm area
- Missing values in installed_capacity_MW
- Missing values in total_project_budget

Example data sources

- Wind Energy Database (thewindpower.net)
- Project developer websites
- Wikipedia

Why it matters

Geographic coordinates (LAT, LON) are a prerequisite for all subsequent API-based enrichment (wind speed, wave height, port distances).

Parsing – Turbine Model & Range Columns

1. Turbine model → turbine_producer + turbine_power_MW

Raw turbine_model	Producer	Power
Siemens Gamesa SWT-7.0-154	Siemens Gamesa	7.0 MW
GE Haliade-X 13 MW	GE	13.0 MW
MHI Vestas V164-8.0	MHI Vestas	8.0 MW
Senvion 6.2M125	Senvion	6.2 MW

- **Producer:** regex matching manufacturers
- **Power:** multiple regex patterns – direct MW, SWT-X.X, SG X.X, V164-X.X; kW→MW conversion; plausible range filter 0.1–20 MW

2. Range columns → min / max / mean

Raw value	Min	Max	Mean
water_depth_m = "20–40"	20	40	30
distance_from_shore_km = "15–25"	15	25	20

Budget Parsing & Inflation Indexation

Budget parsing pipeline

- 1 **Currency detection** – scan for EUR, GBP, USD, NOK, DKK, SEK, PLN
- 2 **Scale detection** – “billion” / “million” keywords
- 3 **Number extraction** – regex, take mean of found values
- 4 **Convert to EUR** – `forex_python` API (historical rates by commissioning year)
- 5 Fallback: static exchange-rate table

Example: “GBP 4.2 billion” (2020)

→ $4.2 \times 10^9 \times \text{rate}(\text{GBP} \rightarrow \text{EUR}, 2020)$

Inflation indexation to 2024

- World Bank CPI API (`wbgapi`)
- Indicator: `FP.CPI.TOTL`
- Country-specific ISO 3 codes (15 EU countries)

$$\text{Budget}_{\text{indexed}} = \text{Budget} \times \frac{\text{CPI}_{2024}}{\text{CPI}_{\text{year}}}$$

- Recent data (≤ 5 years old): no adjustment
- Fallback: static 2% annual inflation

- ❶ **Mean wind speed** – Global Wind Atlas (`wind_stats` library)
 - Weibull parameters (A, k) at hub height **140 m**, roughness 0.03 m
 - Sector-weighted mean: $\bar{v} = \sum_i w_i \cdot A_i \Gamma(1 + \frac{1}{k_i})$
- ❷ **Mean wave height** – Open-Meteo Marine API
 - Endpoint: `marine-api.open-meteo.com/v1/marine`
 - Full year of hourly `wave_height` data per farm location & commissioning year
 - Fallback: current sea-state conditions
- ❸ **Distance from operational port** – EMODnet harbour shapefiles
 - GeoPandas with EPSG:3035 (ETRS89-LAEA) projection
 - Minimum Euclidean distance to nearest port
- ❹ **Distance from construction port** – 29 European ports
 - Esbjerg, Cuxhaven, Bremerhaven, Rotterdam, Hull, Belfast, Gdańsk, ...
 - Same projected-distance algorithm

Data Filtering & Encoding

Mandatory column filter

Rows missing **any** of the following are dropped:

LAT • LON • installed_capacity_MW • total_project_budget

182 rows $\xrightarrow{\text{filter}}$ $\sim$ 170 rows remaining

Additional transformations after filtering

- **OSS responsibility rules** – determined based on country and commissioning year
- **One-hot encoding** of 7 categorical columns:
country, turbine_producer, foundation_type, oss_responsibility, offshore_cable, onshore_connection, model_type

Calculations – Energy Production

Capacity factor CF from wind speed

Wind speed (m/s)	8.5	9.0	9.5	10.0
Capacity factor	0.45	0.50	0.55	0.59

Productivity

$$P_{\text{gross}} = 8760 \times CF \quad [\text{MWh/MW/year}] \quad (1)$$

$$P_{\text{net}} = P_{\text{gross}} \times \left(1 - \underbrace{0.15}_{\text{loss factor}}\right) \quad (2)$$

Annual energy production

$$E_{\text{annual}} = P_{\text{net}} \times \text{Installed Capacity} \quad [\text{MWh}]$$

Unit CAPEX

$$\text{Unit CAPEX} = \frac{\text{Total Budget (EUR)}}{\text{Installed Capacity (MW)}} \quad \left[\frac{\text{EUR}}{\text{MW}} \right]$$

OPEX – distance-based model

Distance bracket	OPEX (kEUR/MW/year)
Base	65
≤ 50 km from port	65 + 5 = 70
50–100 km	65 + 15 = 80
> 100 km	65 + 25 = 90

Capital Recovery Factor

$$CRF = \frac{r}{1 - (1 + r)^{-n}}, \quad r = 8\%, \quad n = \text{project lifetime (years)}$$

Annualised CAPEX

$$CAPEX_{\text{annual}} = \text{Total Budget} \times CRF$$

Levelised Cost of Energy

$$LCOE = \frac{CAPEX_{\text{annual}} + OPEX_{\text{annual}}}{E_{\text{annual}}} \left[\frac{\text{EUR}}{\text{MWh}} \right]$$

Model Training Pipeline

Target variable: Indexed Unit CAPEX

Training steps

- 1 Drop rows with missing target
- 2 Remove outliers (IQR, $1.5\times$ rule)
- 3 Impute missing features (median)
- 4 Standardise features (StandardScaler)
- 5 80/20 train/test split (seed 42)
- 6 5-fold cross-validation
- 7 Select best model by test R^2

Features

- 16 numeric: LAT, LON, area, year, turbine power, water depth (min/max/mean), distance from shore (min/max/mean), port distances, lifetime, wind speed, wave height
- One-hot encoded categoricals: country, turbine producer, foundation type, OSS responsibility, cable, connection, model type

Models & Feature Importance

5 regression models compared

Model	Notes
Ridge Regression	L2 regularisation
LassoCV	L1, $\alpha \in [10^{-3}, 10^3]$ (10 values)
ElasticNetCV	L1/L2 mix, ℓ_1 -ratio $\in \{0.1, 0.5, 0.7, 0.9, 0.95, 0.99, 1.0\}$
Random Forest	Ensemble of decision trees
Gradient Boosting	Sequential boosted trees

Most important features – extracted from best model:

commissioning_year
water_depth_max_m • turbine_power_MW • area_sqkm • mean_wave_height_m

Similarity Analysis – Finding Comparable Farms

9 configurable comparison metrics

- Area (km²)
- Mean water depth (m)
- Mean distance from shore (km)
- Installed capacity (MW)
- Turbine power (MW)
- Distance from port (km)
- Distance from construction port (km)
- Mean wind speed (m/s)
- Mean wave height (m)

3 scaling methods

Z-score (recommended)	Standardise to $\mu = 0, \sigma = 1$
Percentage difference	Relative to proposed farm values
Raw values	No normalisation

Algorithm

$$d = \sqrt{\frac{1}{|M|} \sum_{m \in M} (\tilde{x}_m^{\text{farm}} - \tilde{x}_m^{\text{proposed}})^2} \quad \longrightarrow \quad \text{Similarity} = \frac{1}{1 + d}$$

Interactive Profitability Map

- **Color by** (dropdown):

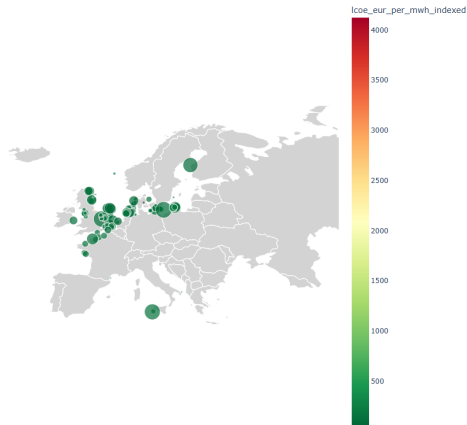
- LCOE (Indexed) [EUR/MWh]
- Unit CAPEX (Indexed) [EUR/MW]
- Annual OPEX [EUR]
- Annual OPEX [EUR/MWh]
- Total Budget (Indexed) [EUR]

- **Marker size:** proportional to installed capacity

- **Hover tooltip:** farm name, country, key metrics, and environmental conditions

- **Geographic scope:** European coast

Profitability Map (color: lcoe_eur_per_mwh_indexed, size: installed capacity)



Prediction Interface – Input Form

14-Parameter Input Form

- 1 LAT, LON (numeric)
- 2 Area [km²]
- 3 Country (dropdown)
- 4 Commissioning year
- 5 Turbine power [MW]
- 6 Turbine producer
- 7 Foundation type
- 8 Water depth min/max [m]
- 9 Distance from shore min/max [km]
- 10 Project lifetime [years]
- 11 Installed capacity [MW]

LAT	56
LON	10
Area (sqkm)	50
Country	Denmark
Commissioning Year	2028
Turbine Power (MW)	10
Turbine Producer	Vestas
Foundation Type	Monopile
Water Depth Min (m)	20
Water Depth Max (m)	40
Distance Shore Min (km)	20
Distance Shore Max (km)	40
Lifetime (years)	25
Installed Capacity (MW)	100

Predict

Prediction Interface

On “Predict” click:

- Auto-enrich via APIs: wind speed, wave height, and port distances
- Assign connection details from country-year rules
- Run best ML model inference
- Compute full profitability metrics

4 Output Cards

- Total Budget (Indexed) [EUR]
- Unit CAPEX (Indexed) [EUR/MW]
- Annual OPEX [EUR]
- LCOE (Indexed) [EUR/MWh]

Prediction Results

Total Project Budget (Indexed)	Unit CAPEX (Indexed)	Annual OPEX	LCOE (Indexed)
2,803,673,164.87 EUR	3,504,591.46 EUR/MW	72,000,000.00 EUR	95.41 EUR/MWh
Distance from Port (shapefile)	Distance from Construction Port		
1,819.3 km	467.8 km		

Prediction results and profitability output cards

Port Similarity Interface

- 1 Configurable list of comparison metrics
- 2 Scaling method dropdown

Output

Table with 5 most similar projects and their characteristics

Closest Historical Farms							
Metrics							
☒ Area (sqkm)							
☒ Mean water depth (m)							
☒ Mean distance from shore (km)							
☒ Installed capacity (MW)							
☐ Turbine power (MW)							
☐ Distance from port (km)							
☐ Distance from construction port (km)							
☐ Mean wind speed (m/s)							
☐ Mean wave height (m)							
Scaling							
Percentage difference							
Percentage difference compares relative deviation from the proposed farm values.							
Farm	Country	Similarity	Distance	Area (sqkm)	Mean water depth (m)	Mean distance from shore (km)	Installed capacity (MW)
Proposed Wind Farm	Denmark	1.000	0.900	80.0	35.0	42.5	800.0
Merkur	Germany	0.749	0.335	47.0	30.0	45.0	396.0
Baltic Eagle	Germany	0.729	0.372	40.0	43.0	30.0	476.0
Berlmer Riffgrund 2	Germany	0.723	0.383	36.0	30.0	54.0	450.0
Nordsee One	Germany	0.717	0.395	41.3	27.5	45.0	332.0
Gode Wind 1	Germany	0.715	0.398	42.0	30.5	33.0	332.0

Port similarity interface and nearest-project comparison table

What was built

- End-to-end pipeline: raw CSV → cleaned & enriched data → profitability calculations → ML models → interactive UI.
- 4 external APIs integrated (Global Wind Atlas, Open-Meteo, World Bank CPI, Forex).
- 5 regression models trained; best selected automatically by R^2 .
- Similarity engine to find comparable historical farms.

Tech stack

Python • Pandas • Scikit-learn • Plotly / Dash • GeoPandas • wind-stats • wbgapi